Developing an AI Agent to Support Administration and Operation of HPC Systems Based on SLURM

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Motivation

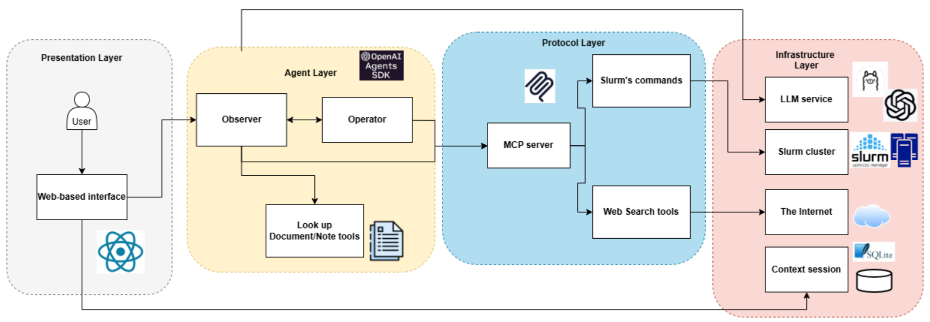
- Slurm CLI is complex — dozens of commands, flags, Bash scripting
- No safety layer for destructive operations (cancel, hold, release)
- Open-weight LLMs lack Slurm tool-calling knowledge
- No standardised benchmark for NL-based Slurm agents

Approach

- Observer/Operator Split: Read-only Observer + state-changing Operator; destructive actions require human confirmation
- Grounded RAG: 273 official Slurm docs → 2,276 chunks; reduces hallucination
- Fine-Tuning: Qwen2.5-14B + QLoRA from GPT-5-mini teacher traces
- Benchmark: 3,135 cases, 11 categories, LLM-as-judge scoring

Implementation

- Stack: React/Vite → FastAPI → MCP → real Slurm cluster
- Tools: 18 typed Slurm tools (~10 Observer, ~8 Operator)
- Training: 3,329 samples · 3 epochs · A40 48 GB · QLoRA 4-bit NF4 · ~$20



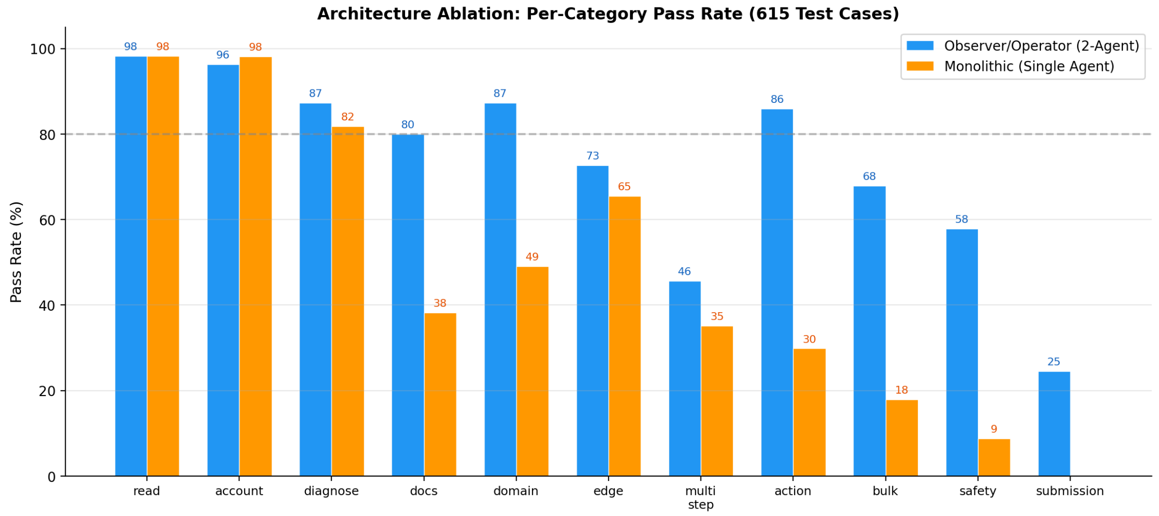
System Architecture

Results — Three-Way Model Comparison

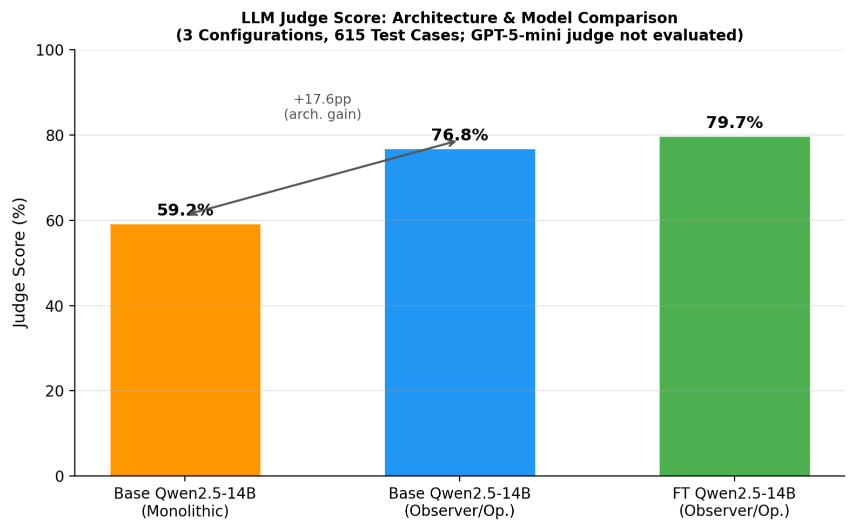
615-case held-out test set (stratified 80/20 split, seed=42)

Model	Pass	Tool	HITL	State	Judge	Lat.
GPT-5-mini	96.6%	98.8%	99.0%	95.9%	74.3%	28s
FT Qwen (ours)	91.4%	89.2%	90.4%	90.0%	79.7%	85s
Base Qwen	72.8%	84.9%	80.8%	87.3%	76.8%	81s
Monolithic ablation	47.2%	77.9%	88.9%	88.9%	59.2%	77s

- FT model closes 78% of the gap to GPT-5-mini baseline
- Observer/Operator beats monolithic by +44.2 pp pass rate
- FT judge (79.7%) exceeds both base Qwen (76.8%) and GPT-5-mini (74.3%)
- Tool partitioning cuts selection confusion: docs +41.8 pp vs. monolithic



Per-Category Results (FT vs Base vs Monolithic)



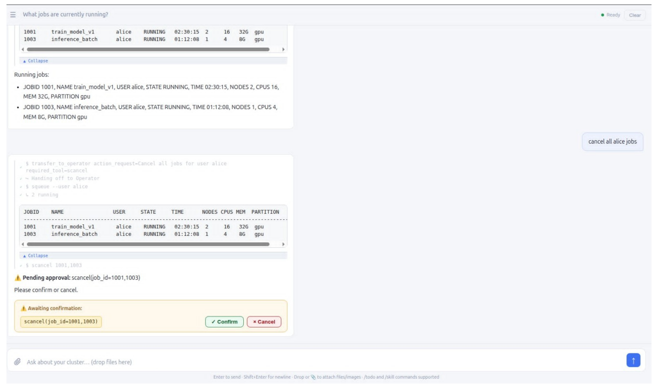
LLM Judge Score Comparison

Contributions

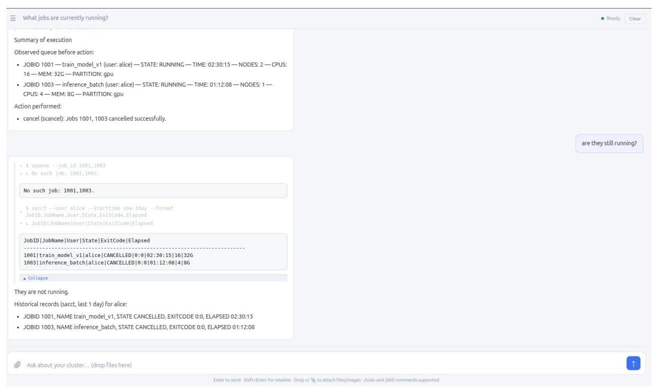
- Novel Observer/Operator dual-agent safety architecture for Slurm
- HITL confirmation workflow for all destructive operations
- Grounded RAG pipeline over 273 official Slurm docs
- 3,135-case curated benchmark with LLM-as-judge scoring
- Cost-effective fine-tuning (~$20) achieving 91.4% pass rate

Future Works

- Multi-tenant authentication, resource isolation, audit logging
- Practitioner user studies for usability validation
- Expand benchmark to cover more Slurm edge cases
- Explore larger open-weight models and multi-turn fine-tuning



HITL Confirmation UI



UI After HITL Confirmation